Jason Bowman

San Francisco, CA • (650) 445-9552 jason@json64.dev sini sinistar

SUMMARY

Staff/Principal-level platform and infrastructure engineer with over a decade architecting distributed systems at hyper-growth scale (Uber, Unity, Highnote), paired with deep programming-language and developer-tooling expertise. Now building *Denful* full-time and in the open: composable, theory-grounded Nix libraries and configuration frameworks built on algebraic effects, scope graphs, and graph rewriting. Equally at home at the bottom of the stack — Kafka, Spanner, Linux, networking — and the top — DSLs, type systems, and effect systems.

EDUCATION

The University of Alabama — Tuscaloosa, AL

Master of Science in Computer Science (MSc), College of Engineering — *2012*

- **Select coursework:** Security · Cryptography · Advanced Computer Networking · Compilers

Bachelor of Science in Computer Science (BSc), College of Engineering — *2011*

- *Minor in Mathematics*

SKILLS

Languages & Platforms: Scala, Nix, Python, Go, Java, C/C++, Ruby, and UNIX shell; 25+ years on GNU/Linux and UNIX; strong functional-programming background
Language Engineering & Tooling: DSL design, type systems, algebraic effects, term and graph rewriting, AST-based codemods and automated migration, macros and metaprogramming; sbt, Bloop, Metals, JMH, Gatling
Distributed Systems: Kafka, stream processing, high-QPS services in hyper-growth environments, elastic compute, API design, performance optimization (zero-allocation hashing, bloom filters), graceful fault detection and recovery
AI & Machine Learning: Reinforcement learning (PPO, SAC, GAIL) and distributed RL training platforms (pre-LLM-boom); LLM agentic workflows, multi-agent orchestration, and spec-driven development
Cloud & Data: GCP and AWS — BigQuery, BigTable, Spanner, Dataflow / Apache Beam / Scio (custom DoFn, Coder, WindowFn), Avro (avro4s), GKE; PostgreSQL, Redis, ZooKeeper, Hadoop, Vertica
Kubernetes & Cloud-Native: Kubernetes (k3s), Cilium (eBPF CNI and network policies), Tailscale, Longhorn distributed storage, microVMs and GPU passthrough
GitOps & Infrastructure-as-Code: ArgoCD, Flux, nixidy, NixOS fleet management (Colmena, nh), declarative secrets (agenix, SOPS), CI/CD and developer-experience tooling
Observability & Reliability: Grafana, Prometheus, Grafana Alloy, metrics and log pipelines, alerting; 24/7 on-call, automated incident detection and remediation, backup and disaster recovery
Security & Compliance: security auditing, applied cryptography (KMS envelope encryption, AES-CBC, key derivation), OIDC and access control, PII/GDPR data governance, HIPAA compliance
Leadership: technical project management, team lead and mentorship, organizational leadership

EXPERIENCE

Denful — Open Source

Co-Founder — *2023 - Present*

- Co-founded an open-source organization building composable, aspect-oriented Nix tooling, grounded in algebraic effects, interaction nets, and graph rewriting, with an active and growing community (*den* at 400+ GitHub stars).
- **gen:** Authored *gen*, the ecosystem's foundational layer: a suite of dependency-light, pure-Nix libraries spanning algebraic primitives, higher-order scope graphs, typed registries, selector and graph-query algebras, and guarded graph rewriting.
- **den:** Lead developer of *den*, a context-aware, aspect-oriented configuration framework that produces NixOS, Darwin, and Home-Manager outputs from a single declarative context, using algebraic effects to make configuration resolution explicit and testable.
- **Domain-Agnostic Configuration:** Generalized the engine beyond NixOS to arbitrary configuration targets, demonstrated by extending it to Kubernetes GitOps through upstream contributions to *nixidy*: a generic object match/rewrite rule engine and CRD-native module accessors.
- **Graph-Based Infrastructure:** Modeled service topology, firewall rules, and access controls as derived properties of a typed relationship graph, synthesizing network policies and ACLs from declared service relationships rather than hand-maintained configuration.

Highnote San Francisco, CA
Senior Software Engineer *2022 - 2023*

- Infrastructure engineering for B2B card issuer fintech, focusing on data systems and large-scale platform migrations.
- **Spanner Replication System:** Implemented transaction log replay and mirroring system using Google Dataflow to enable cross-region Spanner cluster replication, ensuring 99.99% uptime for payment processing systems.
- **Codemods & Migration Tooling:** Built AST-based codemods and automated migration utilities driving mono-repo-wide framework upgrades across 50+ microservices, cutting manual effort ~80% and eliminating version skew.
- Built data pipeline infrastructure for real-time payment analytics and fraud detection, processing millions of transactions daily with sub-100ms latency requirements.

Unity Technologies San Francisco, CA
Senior Software Engineer / Tech Lead *2018 - 2022*

- Led engineering teams and architected large-scale data platform infrastructure serving analytics, advertising, and ML workloads at one of the world's largest mobile ad networks.
- **Data Platform (Sole Architect):** Sole architect and primary engineer of Unity's company-wide event-ingestion platform — the production replacement for a legacy daily-batch flow, built during the AWS-to-GCP migration while leading a team of 6. On Apache Beam/Scio and Dataflow (Kafka → schema'd Avro → GCS → BigQuery), it sustained 1M events/sec (10B/day) and cut end-to-end data-availability latency from 24+ hours to under 15 minutes at 99.99% availability, consolidating three previously independent ingestion pipelines (analytics, advertising, CDP) into one platform adopted company-wide.
- **GDPR & CCPA Compliance:** Architected the platform's privacy layer and led the cross-functional rollout with Legal and company-wide data owners. PII was split into a separate KMS-encrypted stream and access-controlled store, with subject-level cryptographic compliance keys enabling right-to-erasure across the data lake — plus PII hashing, geo-coordinate fuzzing, GeoIP enrichment, and automated statistical PII classification — flagging fields by value cardinality and uniqueness for data-owner review and refinement. Per-owner bucketing provided tenant-scoped access controls and cost chargeback.
- **Runtime Control Plane:** Designed a live control plane — BigTable-backed schema, routing, and encoding registries hot-reloaded into running pipelines — so data owners onboarded new event types and evolved schemas with zero redeploys, decoupling platform operations from per-team data changes. Automated schema derivation inferred Avro contracts from existing data.

- **Reliability Tier:** Built the platform's fault-tolerance layer — dead-letter capture/replay with rate-limited backoff and BigQuery load-job monitoring with automatic retry — sustaining delivery guarantees at 10B events/day.
- **ML-Agents RL Platform:** Architected a cloud-native reinforcement-learning platform on GKE for Unity's ML-Agents — applied RL (PPO, SAC, GAIL) years before the current AI boom — orchestrating distributed training experiments and Unity simulation builds, and scaling researchers' RL training across hundreds of nodes.
- **Unity Simulation & Robotics:** Built remote tunneling infrastructure allowing secure access to Unity instances in shared cloud environments. Developed tooling to bridge on-premise robotics workflows with cloud-based simulation infrastructure.

Uber Technologies, Inc San Francisco, CA
Software Engineer *2015 - 2018*

Marketplace Engineering // Fares *2016 - 2018*

- Led development of end-user facing product features spanning back-end services and mobile applications.
- **[Tech Lead] Pricing Integrity Engine:** Built real-time and batch data validation platform for fare data with configurable business rules, analytics, and automatic remediation.
- **Fares Platform:** Developed next-generation fare calculation platform for all Uber products using descriptive DSL, integrating tolls, maps, surge, and subscription data.
- **Promotions Platform:** Built auditing and remediation tools for post-transaction analysis of user promotions across real-time systems.

Site Reliability Engineering // Streaming SRE *2015 - 2016*

- Senior engineer on newly formed Streaming SRE team. Led team building and mentored junior engineers while scaling Kafka infrastructure to thousands of brokers processing >500GiB/s across multiple regions.
- **Kafka Infrastructure Projects:** Built an intelligent partition-redistribution engine spanning hundreds of brokers across many clusters, driving load-distribution standard deviation from ∼53% to 0.0001% of mean, eliminating disk hotspots and self-healing broker failures. Filesystem and configuration tuning delivered over 5× (500%) throughput in synthetic benchmarks; in production, p99 produce latency dropped from recurring ∼500ms spikes to a stable ∼10ms (∼50×) and request-handler utilization fell from ∼80% to ∼20% — averting a projected nine-figure migration to an all-SSD broker fleet by proving the existing disk stack could meet I/O targets through optimization alone.

Platform Engineering // Data Engineering Embed *2015*

- Supported Data organization representing 30% of Uber's hardware footprint. Built automation for provisioning, replication, ETL, and cluster management across Vertica, Hadoop, Redis, and MemSQL.
- Designed data security architecture and compliance tools for global expansion into China, India, and Russia.
- **[Tech Lead] uServer:** Built self-service hardware provisioning platform adopted company-wide, provisioning hundreds of thousands of bare-metal hosts across our datacenters.

Rocket Lawyer San Francisco, CA
Senior Systems Engineer (original title: Systems Engineer) *2013 - 2015*

- Designed service infrastructure and provided 24/7 on-call support for a high volume e-commerce site in the legal services sector, running primarily on the JVM and a legacy .NET platform. Improved site reliability from 99% uptime to 99.99% and reduced p99 latencies from 2000ms to 150ms.
- Performance analysis and optimization of application services and supporting infrastructure services.

- Extended application code to expose internal application state, key performance metrics, enrich logging, and reduce overall latencies. Implemented dynamic service discovery and reconfiguration via ZooKeeper.
- Built tooling for continuous integration, dynamic configuration management, deployment management, monitoring, and automated incident detection and remediation.

nine.is Tuscaloosa, AL
Software Engineer / Web Developer *2012*

- Full-stack development and team leadership for design company transitioning to in-house software development. Built applications using PHP, JavaScript, and Ruby on Rails.

The University of Alabama Tuscaloosa, AL
Graduate Research Assistant *2010 - 2012*

- Research on network optimization and security for next-generation internet infrastructure. Developed custom TCP variants and performed security analysis for GENI experimental network testbed.

Brewer Porch Children's Center Tuscaloosa, AL
Unix Systems Administrator *2004 - 2006*

- Systems administration for healthcare organization. Managed Linux/Solaris infrastructure and ensured HIPAA compliance for patient data systems.

PUBLICATIONS

- Dawei Li, Jason Bowman, Xiaoyan Hong "Evaluation of Security Vulnerabilities by Using ProtoGENI as a Launchpad", IEEE Globecom 2011, Houston, USA, Dec. 2011.
- M. Anderson, P. Kilgo, and J. Bowman. "RDIS: Generalizing domain concepts to specify device to framework mappings." In International Conference on Robotics and Automation, 2012. http://ua-robotics.net/index.php?title=RDIS

OUTREACH AND COMMUNITY INVOLVEMENT

- Mentor for Hackbright Academy, a 12 week intensive coding bootcamp focused on empowering women aiming to pursue a career in the software engineering field.
- Mentor for dev/Mission, a non-profit program targeting low income youth from 16-24 to expose them to potential careers in technology.
- Organized and ran an introduction to Scala and Functional Programming course based on Martin Oderskyś course and book. Resources available on my personal github.
- Served as president from 2010 - 2012 of the Association for Computing Machinery student chapter at The University of Alabama.
- Organized a mentoring/tutoring program for freshman and sophomore computer science students in an effort to improve program retention and graduation rates. Still going to this day, and graduation rates of freshman classes are up from 5% to nearly 50%.

INTERESTS

- **Professional:** GNU/Linux · Distributed Systems · Open Source · System Architecture · Operating Systems · Functional Programming · Stream Processing · Virtualization · Generative Programming · Language Development
- **Personal:** Privacy · Reverse Engineering · Game Development & Theory · Music Production · Electrical Engineering · Embedded Systems · Fabrication and DIY · Robotics · Mechanical Keyboards · CAD Design · Rock Climbing · Emacs